DATABASE RELATIONSHIP ANALYSIS USING JOINS

INTRODUCTION

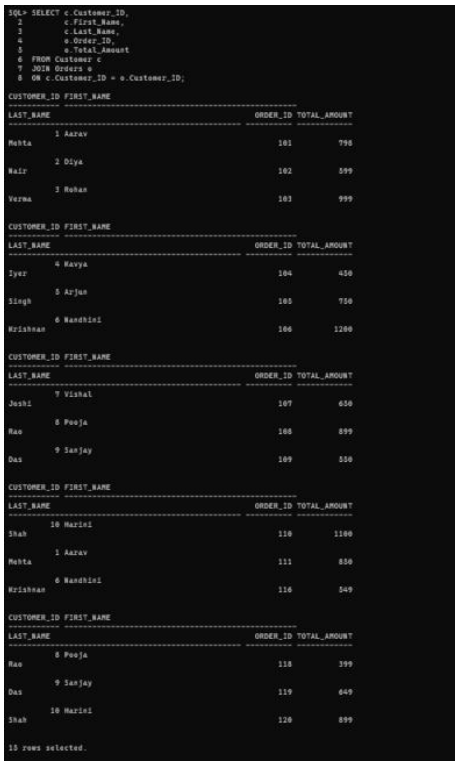
SQL JOIN operations are used to retrieve and combine related data from multiple tables in a database. In the Vilvah E-Commerce Database, JOINs connect customer, product, order, order details, and payment information. Operations such as INNER JOIN, LEFT JOIN, and RIGHT JOIN help retrieve customer purchase history, display complete order details, and generate payment transaction reports. These queries improve data analysis and support effective management of customer orders and business activities.

Objectives

1. Combine Customer, Product, Orders, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. Combining Multiple Tables

The Vilvah database contains different tables that store customer, product, order, and payment information. JOIN operations combine related records from these tables to retrieve useful information.



```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         o.Order_ID,
  5         o.Total_Amount
  6  FROM Customer c
  7  JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID TOTAL_AMOUNT
-------------------------------------------------- -------- ------------
          1 Aarav
Mehta                                                   101          798

          2 Diya
Nair                                                    102          599

          3 Rohan
Verma                                                   103          999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID TOTAL_AMOUNT
-------------------------------------------------- -------- ------------
          4 Kavya
Iyer                                                    104          450

          5 Arjun
Singh                                                   105          750

          6 Nandhini
Krishnan                                                106         1200

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID TOTAL_AMOUNT
-------------------------------------------------- -------- ------------
          7 Vishal
Joshi                                                   107          630

          8 Pooja
Rao                                                     108          899

          9 Sanjay
Das                                                     109          530

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID TOTAL_AMOUNT
-------------------------------------------------- -------- ------------
         10 Harini
Shah                                                    110         1100

          1 Aarav
Mehta                                                   111          850

          6 Nandhini
Krishnan                                                116          549

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID TOTAL_AMOUNT
-------------------------------------------------- -------- ------------
          8 Pooja
Rao                                                     118          399

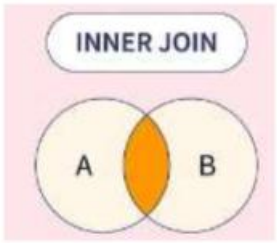
          9 Sanjay
Das                                                     119          649

         10 Harini
Shah                                                    120          899

15 rows selected.
```

2. INNER JOIN

INNER JOIN returns only records that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         o.Order_ID,
  5         o.Order_Date,
  6         o.Total_Amount
  7  FROM Customer c
  8  INNER JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------
          1 Aarav
Mehta                                                   101 20-SEP-26
         700

          2 Diya
Nair                                                    102 21-SEP-26
         599

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------

          3 Rohan
Verma                                                   103 22-SEP-26
         999

          4 Navya
Iyer                                                    104 23-SEP-26

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------
         450

          5 Arjun
Singh                                                   105 24-SEP-26
         750

          6 Nandhini

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------
Krishnan                                                106 25-SEP-26
        1200

          7 Vishal
Joshi                                                   107 26-SEP-26
         650


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------
          8 Pooja
Rao                                                     108 27-SEP-26
         899

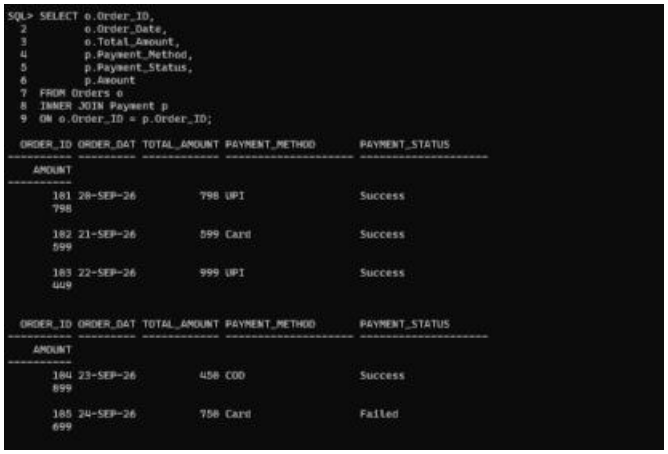
          9 Sanjay
Das                                                     109 28-SEP-26
         550

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- -------- --------
TOTAL_AMOUNT
------------

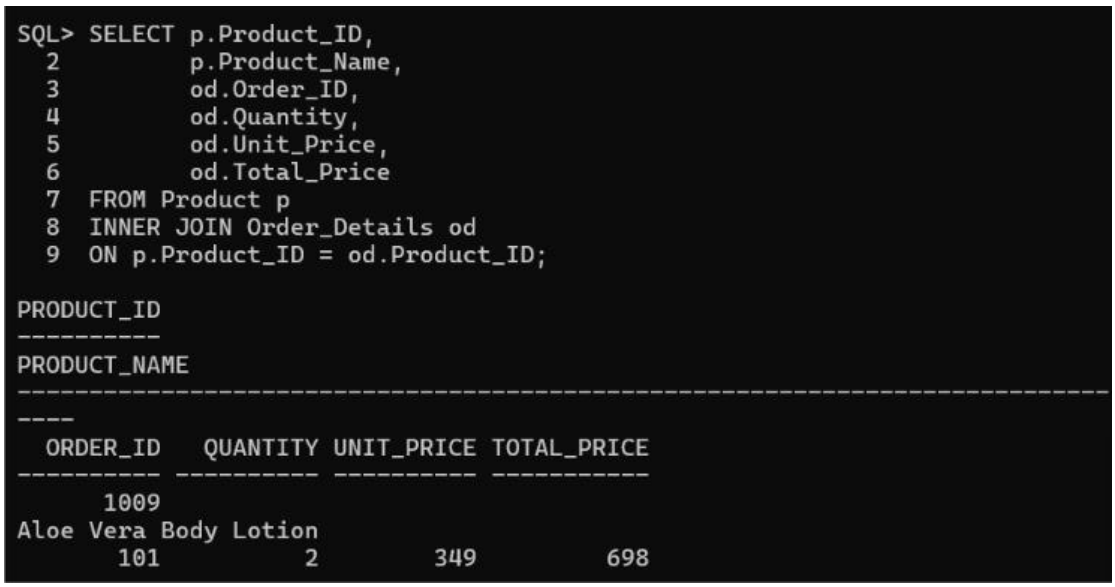
         10 Harini
Shah                                                    110 29-SEP-26
        1100
```



2.2 Orders and Payment

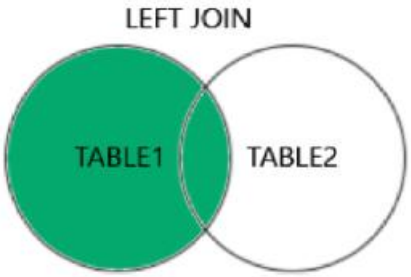


2.3 Product and Order_Details

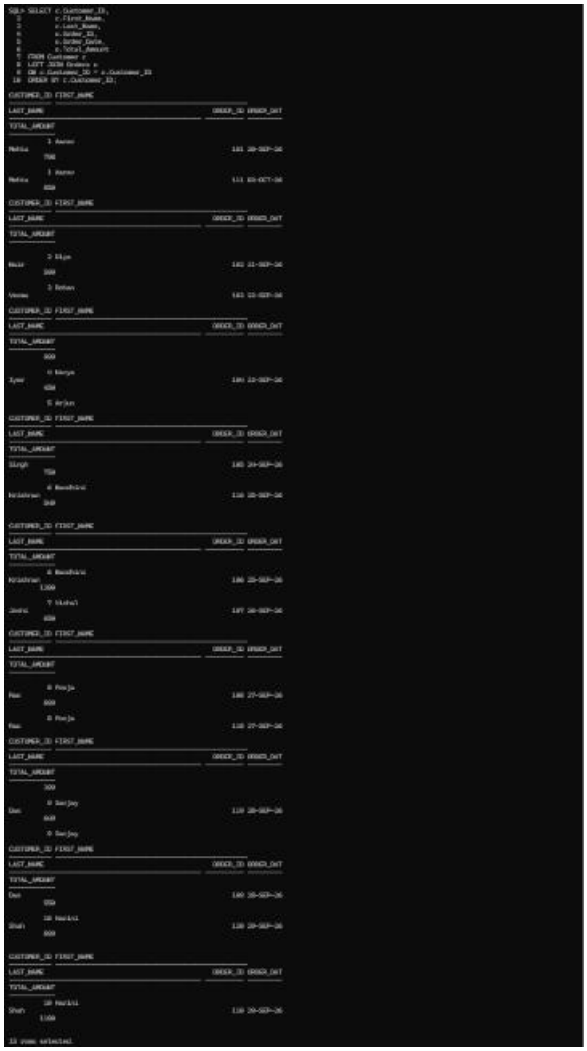


3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, the right-table columns show NULL.

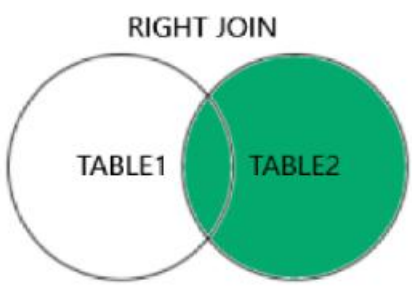


3.1 Display All Customers and Their Orders

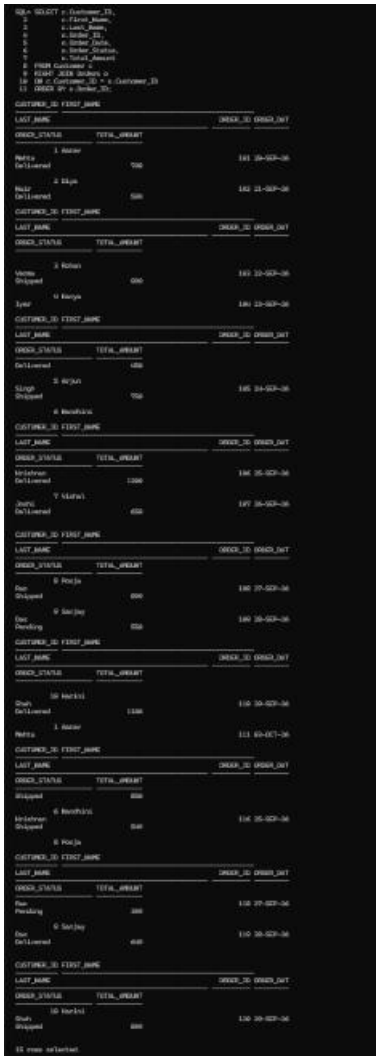


4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details



5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables to display comprehensive order information.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         o.Order_ID,
  5         o.Order_Date,
  6         p.Product_Name,
  7         od.Quantity,
  8         od.Unit_Price,
  9         od.Discount,
 10         od.Total_Price,
 11         pay.Payment_Method,
 12         pay.Payment_Status
 13  FROM Customer c
 14  JOIN Orders o
 15  ON c.Customer_ID = o.Customer_ID
 16  JOIN Order_Details od
 17  ON o.Order_ID = od.Order_ID
 18  JOIN Product p
 19  ON od.Product_ID = p.Product_ID
 20  LEFT JOIN Payment pay
 21  ON o.Order_ID = pay.Order_ID
 22  ORDER BY o.Order_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   DISCOUNT TOTAL_PRICE PAYMENT_METHOD
---------- ---------- ---------- ----------- --------------------
PAYMENT_STATUS
--------------------
         1 Aarav
Mehta                                                 101 20-SEP-26
Aloe Vera Body Lotion

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   DISCOUNT TOTAL_PRICE PAYMENT_METHOD
---------- ---------- ---------- ----------- --------------------
PAYMENT_STATUS
--------------------
         2        349          0         698 UPI
Success
```

6. Customer Purchase History

This query displays products purchased by each customer.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         o.Order_ID,
  5         o.Order_Date,
  6         p.Product_Name,
  7         od.Quantity,
  8         od.Total_Price
  9  FROM Customer c
 10  JOIN Orders o
 11  ON c.Customer_ID = o.Customer_ID
 12  JOIN Order_Details od
 13  ON o.Order_ID = od.Order_ID
 14  JOIN Product p
 15  ON od.Product_ID = p.Product_ID
 16  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_PRICE
---------- -----------
         1 Aarav
Mehta                                                 101 20-SEP-26
Aloe Vera Body Lotion
         2         698
```

7. Customer Purchase History with Payment

This query combines customer purchase history with payment information.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         o.Order_ID,
  5         p.Product_Name,
  6         od.Quantity,
  7         o.Total_Amount,
  8         pay.Payment_Method,
  9         pay.Payment_Status
 10  FROM Customer c
 11  JOIN Orders o
 12  ON c.Customer_ID = o.Customer_ID
 13  JOIN Order_Details od
 14  ON o.Order_ID = od.Order_ID
 15  JOIN Product p
 16  ON od.Product_ID = p.Product_ID
 17  LEFT JOIN Payment pay
 18  ON o.Order_ID = pay.Order_ID
 19  ORDER BY c.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------------
LAST_NAME                                                ORDER_ID
-------------------------------------------------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY TOTAL_AMOUNT PAYMENT_METHOD       PAYMENT_STATUS
---------- ------------ -------------------- --------------------

         1 Aarav
Mehta                                                    101
Aloe Vera Body Lotion
         2          798 UPI                  Success
```

8. Multi-Table Business Report

The following query generates a customer purchase report with product and order information.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name,
  3         c.Last_Name,
  4         COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  5         SUM(od.Quantity) AS Total_Quantity,
  6         SUM(od.Total_Price) AS Total_Product_Value
  7  FROM Customer c
  8  JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID
 10  JOIN Order_Details od
 11  ON o.Order_ID = od.Order_ID
 12  GROUP BY c.Customer_ID,
 13           c.First_Name,
 14           c.Last_Name
 15  ORDER BY Total_Product_Value DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------

         1 Aarav
Mehta                                                         1              4
            1496

         6 Nandhini
Krishnan                                                      1              3
            1200

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------

        10 Harini
Shah                                                          1              2
            1100

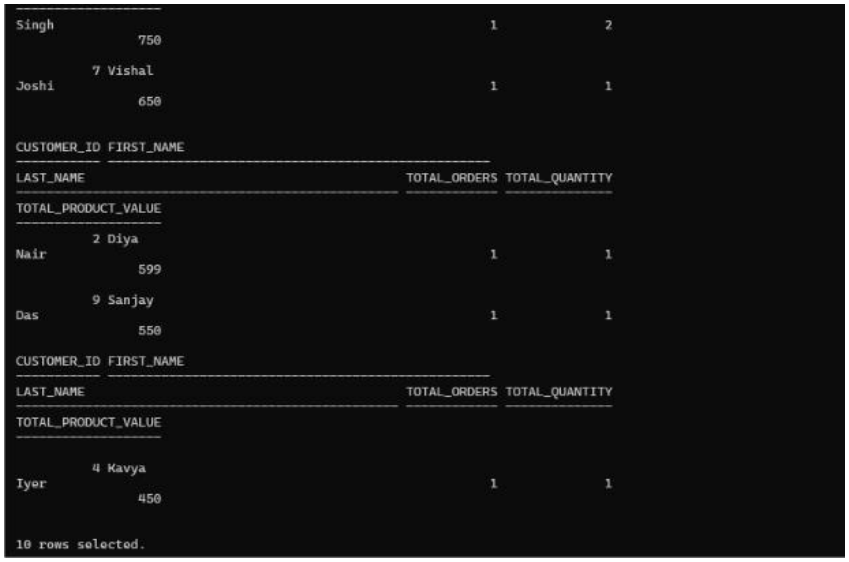
         3 Rohan
Verma                                                         1              2

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------
             999

         8 Pooja
Rao                                                           1              2
             899

         5 Arjun

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------
Singh                                                         1              2
```



```
--------------------
Singh                                                    1              2
                750

         7 Vishal
Joshi                                                    1              1
                650


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------
         2 Diya
Nair                                                     1              1
                599

         9 Sanjay
Das                                                      1              1
                550

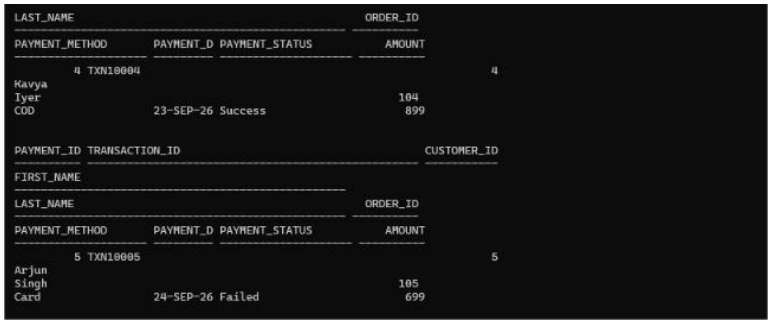
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_ORDERS TOTAL_QUANTITY
-------------------------------------------------- ------------ --------------
TOTAL_PRODUCT_VALUE
-------------------

         4 Kavya
Iyer                                                     1              1
                450


10 rows selected.
```

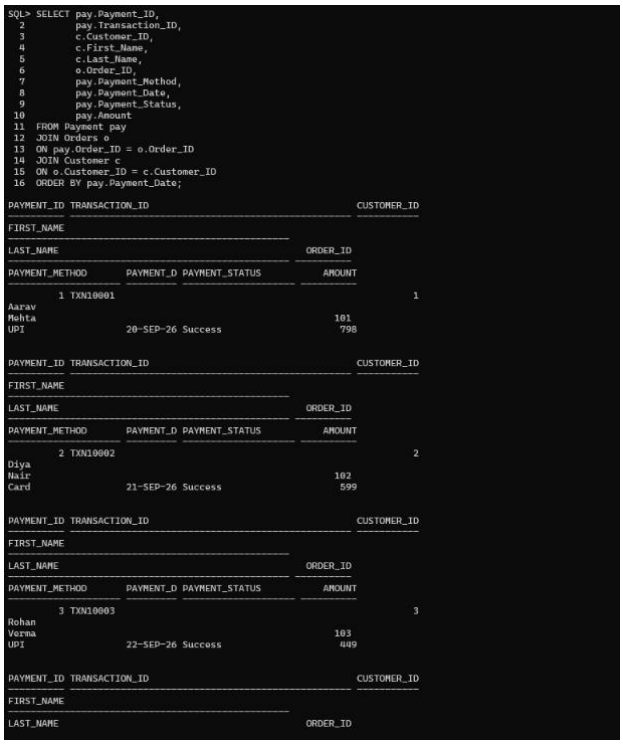
9. Payment Transaction Report

This query displays payment transactions along with customer and order details.



```
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PAYMENT_METHOD       PAYMENT_D PAYMENT_STATUS          AMOUNT
-------------------- --------- -------------------- ----------
         4 TXN10004                                           4
Kavya
Iyer                                                       104
COD                  23-SEP-26 Success                     899

PAYMENT_ID TRANSACTION_ID                                  CUSTOMER_ID
---------- --------------------------------------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PAYMENT_METHOD       PAYMENT_D PAYMENT_STATUS          AMOUNT
-------------------- --------- -------------------- ----------
         5 TXN10005                                           5
Arjun
Singh                                                      105
Card                 24-SEP-26 Failed                      699
```



```
SQL> SELECT pay.Payment_ID,
  2         pay.Transaction_ID,
  3         c.Customer_ID,
  4         c.First_Name,
  5         c.Last_Name,
  6         o.Order_ID,
  7         pay.Payment_Method,
  8         pay.Payment_Date,
  9         pay.Payment_Status,
 10         pay.Amount
 11  FROM Payment pay
 12  JOIN Orders o
 13  ON pay.Order_ID = o.Order_ID
 14  JOIN Customer c
 15  ON o.Customer_ID = c.Customer_ID
 16  ORDER BY pay.Payment_Date;

PAYMENT_ID TRANSACTION_ID                                  CUSTOMER_ID
---------- --------------------------------------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PAYMENT_METHOD       PAYMENT_D PAYMENT_STATUS          AMOUNT
-------------------- --------- -------------------- ----------
         1 TXN10001                                           1
Aarav
Mehta                                                      101
UPI                  20-SEP-26 Success                     798

PAYMENT_ID TRANSACTION_ID                                  CUSTOMER_ID
---------- --------------------------------------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PAYMENT_METHOD       PAYMENT_D PAYMENT_STATUS          AMOUNT
-------------------- --------- -------------------- ----------
         2 TXN10002                                           2
Diya
Nair                                                       102
Card                 21-SEP-26 Success                     599

PAYMENT_ID TRANSACTION_ID                                  CUSTOMER_ID
---------- --------------------------------------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PAYMENT_METHOD       PAYMENT_D PAYMENT_STATUS          AMOUNT
-------------------- --------- -------------------- ----------
         3 TXN10003                                           3
Rohan
Verma                                                      103
UPI                  22-SEP-26 Success                     449

PAYMENT_ID TRANSACTION_ID                                  CUSTOMER_ID
---------- --------------------------------------------------- -----------
FIRST_NAME
--------------------------------------------------
LAST_NAME                                          ORDER_ID
```